

END SEMESTER ASSESSMENT (ESA) - Jan - May 2024**UE22CS242A - Web Technologies****Total Marks : 100.0**

1.a. What is HTTP? Explain the structure of HTTP request message. (6.0 Marks)

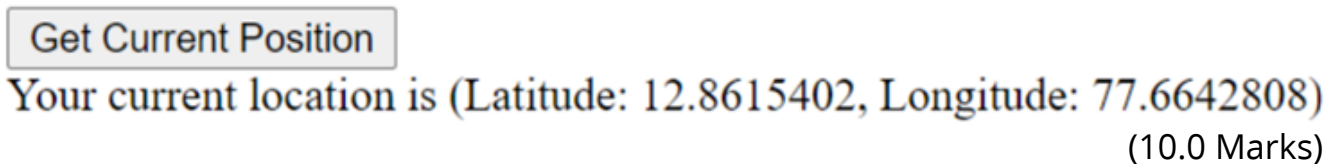
1.b. Create a Placement form with the following controls:

- a. A text box to collect the student's name and SRN
- b. A multiline input field to collect college name and address
- c. A student should give choice to tick their department from a list of departments
 - i. CSE
 - ii. ECE
 - iii. MECH
- d. A collection of three radio buttons for Semester option that are labelled as follows:
 - i. IV
 - ii. V
 - iii. VI
- e. Submit and reset Button (10.0 Marks)

1.c. Describe the structure of HTTP Request and Response message using an example. (5.0 Marks)

1.d. With a neat diagram, explain the CSS Box model and its significance. (4.0 Marks)

2.a. What is Geo Location? Explain the two methods of navigator.geolocation object?
Write a program to display your current location on the browser when the button "Get Current Position" is clicked as shown below.



(10.0 Marks)

2.b. What is AJAX? Explain in detail any 3 XHR object properties. (6.0 Marks)

2.c. Give the difference between JSON and XML.

Represent the details of three 6th sem students both in JSON and XML. Consider the attributes id, name, branch, CGPA (9.0 Marks)

3.a. Create a simple server (use http request GET method) that will allow you to display an HTML file on the browser given the pathname. Set default pathname to index.html. Handle errors by displaying file not found message (10.0 Marks)

3.b. With a neat diagram explain component life cycle. (8.0 Marks)

3.c. What is the significance of key property? Describe with an example. (7.0 Marks)

4.a. What is RESTful API? Explain any 4 design specification/constraints of REST API. (6.0 Marks)

4.b. Write server-side script in JavaScript to route requests for GET and POST requests for flight details. The details are stored in the mongodb database in the following format.

```
{from:"BLR", to:"DEL", dept:"12:25", arrv:"14:25", flnum:"6E-2428"}
```

The server script should support the following routes:

- GET /flights – return details of all flights
- GET /flights/:from/:to – return details of flight between specific airports
- POST /flights – save details of a flight and return a success or error message

(10.0 Marks)

4.c. Explain the role of express middleware function. What are the different types of middleware an express application can use? (4.0 Marks)

4.d. Write a program to upload a file to Node.js server using express file upload library. (5.0 Marks)